Operation Manual

Thorlabs Instrumentation

LED Assemblies

Model Numbers

M365L1	M530L1	LEDC1
M385L1	M590L1	.
M405L1	M627L1	LEDC28
M455L1	M660L1	LEDC33
M470L1	M850L1	.
M505L1		LEDC52

2009

Revision History

Revison	Date	Summary
1	01.04.09	Initial Issue
2	07.07.09	MxLED Series added
3	25.09.09	Series Harmonization

Safety

Attention

All statements regarding safety of operation and technical data in the instruction manual will only apply when the unit is operated correctly.
The mounted LED of the MxxxLx series must not be operated in explosion endangered environments!
The mounted LED of the MxxxLx series must only be connected with duly shielded connection cables.
Only with written consent from Thorlabs may changes to single components be carried out or components not supplied by Thorlabs be used.
Do not remove covers!
Refer servicing to qualified personal!

Warnings

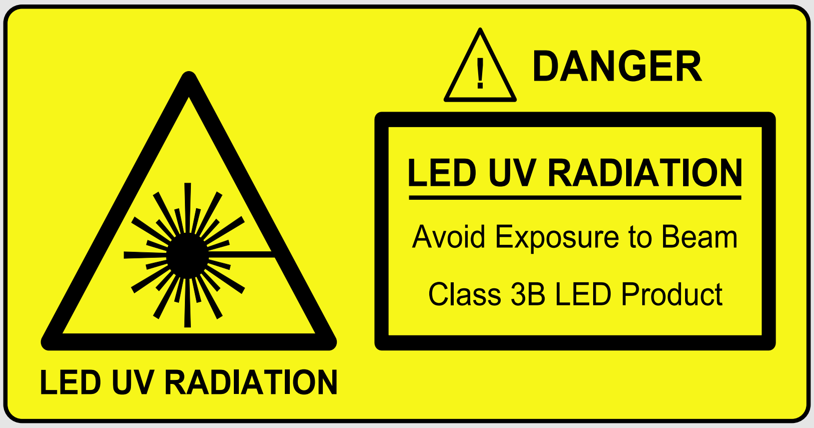
Inappropriate use of any Thorlabs High Power LED product could result in permanent eye damage.

To prevent injury, these products must be used in accordance with the International Standard “Photobiological Safety of Lamps & Lamp Systems” CEI IEC 62471.

When Thorlabs high power LED’s are used in microscope applications as a replacement for mercury Vapor lamps, the same precautions should be taken as those applying to mercury Vapor lamps.

When Thorlabs High Power LED’s are used in other applications, they should be used in accordance with CEI IEC 62471.

Warnings



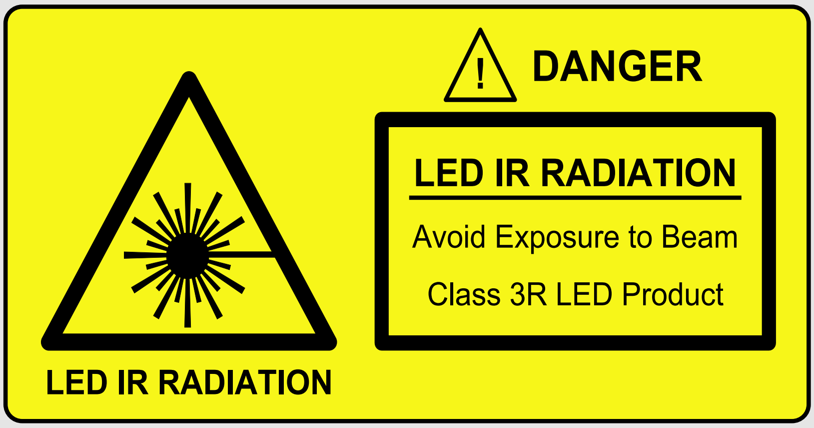
The M365L1, the M385L1 and the M405L1 contain an UV light LED! The LED radiates intense UV light during operation. Precautions must be taken to prevent looking directly at the UV light with unprotected eyes.

Do not look directly into the UV light or look through the optical system during operation of the device. This can be harmful to the eyes even for brief periods due to the high intensity of UV light.

If there is a possibility of reflection of the UV light, UV light protective glasses must be used to prevent the UV light entering the eye.

If viewing the UV light is necessary, UV light protective glasses must be worn to avoid eye damage by the UV light.

Warnings



The M850L1 contains an IR light LED! The LED radiates intense IR light during operation. Precautions must be taken to prevent looking directly at the IR light with unprotected eyes.

Do not look directly into the IR light or look through the optical system during operation of the device. This can be harmful to the eyes even for brief periods due to the high intensity of IR light.

If there is a possibility of reflection of the IR light, IR light protective glasses must be used to prevent the IR light entering the eye.

If viewing the IR light is necessary, IR light protective glasses must be worn to avoid eye damage by the IR light.

Cautions

During normal operations, the casing temperature of Thorlabs High Power LED products will rise by up to 25°C (45°F) above ambient temperature.

To prevent higher case temperatures, the products should be operated without hindrance to air movement surrounding the convective cooling fins.

Table of Content

1 General Information

Developments in the design of Light Emitting Diodes (LEDs) have resulted in devices that have significantly more power and can rival certain halogen lamps and gas discharge lamps. Power levels are now high enough for LEDs to present a practical alternative to these traditional light sources. They offer several advantages, including better stability, longer life, lower cost of replacement and lower risk (no glass bulbs to break).

All of Thorlabs mounted & heatsinked uncollimated LED assemblies are compatible with our range of LED Collimator assemblies and can mate to the standard and epi-illumination ports on most readily available commercial microscopes. A range of different colors (dominant wavelengths) are available, making these devices applicable for wide field fluorescence microscopy when combined with suitable excitation and emission filters.

Each LED module has a flying lead terminated in a 4 pin M8 plug, compatible with Thorlabs series of LED drivers.

2 Getting Started

2.1 Unpacking

Inspect the shipping container for damage.

If the shipping container seems to be damaged, keep it until you have inspected the contents and you have inspected the MxxxLx mechanically and electrically.

Verify that you have received the following items:

- MxxxLx Mounted LED assembly
- Operation Manual

2.2 Preparation

Before you connect tie LED to a LED driver, check if the LED can handle the maximum LED current or the LED current limit is configured correctly. The MxxxLx features an EEPROM which contains the maximum LED current, wavelength and forward voltage of the LED.

1. Connect the MxxxLx to the LED driver.
2. Check the current limit.
3. Switch on the LED driver.

2.3 Physical Overview

Figure 1 Mounted LED Assembly

The following picture shows the male connector of the MxxxLx mounted LED assembly. It is a standard M8x1 sensor circular connector. Pin 1 and 2 are the connection to the LED. Pin 3 and 4 are used for the internal EEPROM in the MxxxLx series. Do not use these connections when using a different LED driver than Thorlabs series of LED drivers.

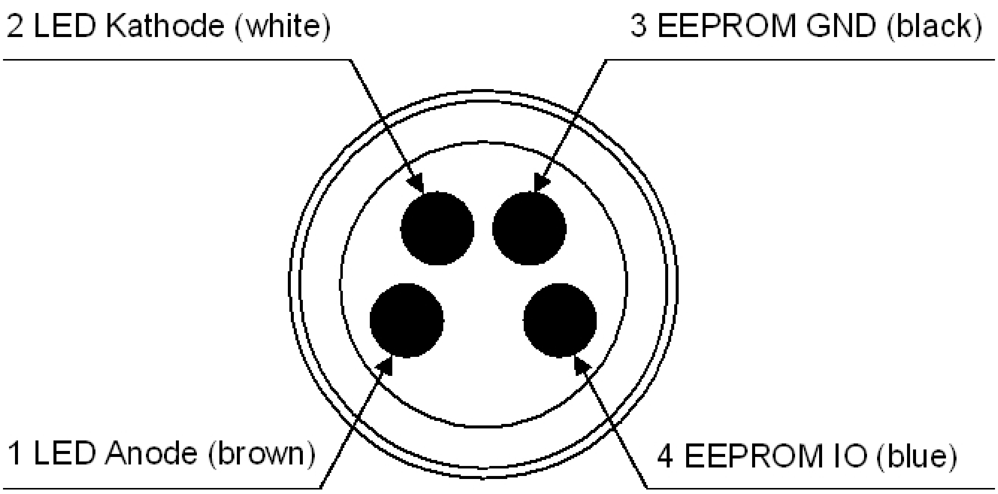


Figure 2 Male Connector of the MxxxLx

The LEDC series consists of the MxxxLx mounted LED and a collimation assembly compatible for one of the following microscopes:

- Olympus BX and IX microscopes
- Leica DMI microscopes

- Nikon Eclipse (F-Mount) Microscopes
- Zeiss Axioskop Microscopes

	Olympus	Leica	Nikon	Zeiss
365nm	LEDC33	LEDC34	LEDC35	LEDC36
385nm	LEDC37	LEDC38	LEDC39	LEDC40
405nm	LEDC41	LEDC42	LEDC43	LEDC44
455nm	LEDC1	LEDC2	LEDC3	LEDC4
470nm	LEDC5	LEDC6	LEDC7	LEDC8
505nm	LEDC9	LEDC10	LEDC11	LEDC12
530nm	LEDC13	LEDC14	LEDC15	LEDC16
590nm	LEDC21	LEDC22	LEDC23	LEDC24
627nm	LEDC25	LEDC26	LEDC27	LEDC28
660nm	LEDC45	LEDC46	LEDC47	LEDC48
850nm	LEDC49	LEDC50	LEDC51	LEDC52
Cold White	LEDC17	LEDC18	LEDC19	LEDC20

3 Operating the MxxxLx

3.1 Thorlabs LED Driver

Thorlabs offers a wide range of LED drivers. The following table gives an overview about the different LED drivers.

Model	Maximum Current	Maximum Forward Voltage	Number of Channels	Current Limit Capability	Notes
LEDD1	700 mA	13 V	1	No	Simple LED Driver
LEDD1A	1000 mA	10 V	1	No	Simple LED Driver
DC2100	2000 mA	24 V	1	Yes	High Power LED Driver
DC4100	1000 mA	5 V	4	Yes	4 Channel LED Driver
DC2500	2000 mA	24 V	1	Yes	High Power LED Driver with Temperature Control

4 Specifications

4.1 M365L1

Wavelength (nominal)	365nm
Optical Beam Power (LED Output)	350mW
Optical beam power (Collimated Beam)	110mW
Maximum LED Current	700mA
LED Forward Voltage (nominal)	4.4V
Lifetime	500 hours

4.2 M385L1

Wavelength (nominal)	385nm
Optical Beam Power (LED Output)	450mW
Optical beam power (Collimated Beam)	140mW
Maximum LED Current	700mA
LED Forward Voltage (nominal)	4.3V
Lifetime	500 hours

4.3 M405L1

Wavelength (nominal)	405nm
Optical Beam Power (LED Output)	670mW
Optical beam power (Collimated Beam)	370mW
Maximum LED Current	1000mA
LED Forward Voltage (nominal)	4.6V
Lifetime[1)]	100,000 hours

[1)] 90% Radiant Flux Maintenance on constant current of 700mA and with junction temperature at or below 115°C.

4.4 M455L1

Wavelength (nominal)	455nm
Optical Beam Power (LED Output)	700mW
Optical beam power (Collimated Beam)	160mW
Maximum LED Current	700mA
LED Forward Voltage (nominal)	6.8V
Lifetime	100,000 hours

4.5 M470L1

Wavelength (nominal)	470nm
Optical Beam Power (LED Output)	625mW
Optical beam power (Collimated Beam)	195mW
Maximum LED Current	700mA
LED Forward Voltage (nominal)	6.8V
Lifetime	100,000 hours

4.6 M505L1

Wavelength (nominal)	505nm
Optical Beam Power (LED Output)	420mW
Optical beam power (Collimated Beam)	115mW
Maximum LED Current	700mA
LED Forward Voltage (nominal)	6.8V
Lifetime	100,000 hours

4.7 M530L1

Wavelength (nominal)	530nm
Optical Beam Power (LED Output)	275mW
Optical beam power (Collimated Beam)	67mW
Maximum LED Current	700mA
LED Forward Voltage (nominal)	6.8V
Lifetime	100,000 hours

4.8 M590L1

Wavelength (nominal)	590nm
Optical Beam Power (LED Output)	150mW
Optical beam power (Collimated Beam)	36mW
Maximum LED Current	1400mA
LED Forward Voltage (nominal)	3.4V
Lifetime	100,000 hours

4.9 M627L1

Wavelength (nominal)	625nm
Optical Beam Power (LED Output)	500mW
Optical beam power (Collimated Beam)	174mW
Maximum LED Current	1400mA
LED Forward Voltage (nominal)	3.4V
Lifetime	100,000 hours

4.10 M660L1

Wavelength (nominal)	660nm
Optical Beam Power (LED Output)	850mW
Optical beam power (Collimated Beam)	390mW
Maximum LED Current	1500mA
LED Forward Voltage (nominal)	3.4V
Lifetime[1)]	100,000 hours

[1)] 90% Radiant Flux Maintenance on constant current of 1000mA and with junction temperature at or below 110°C.

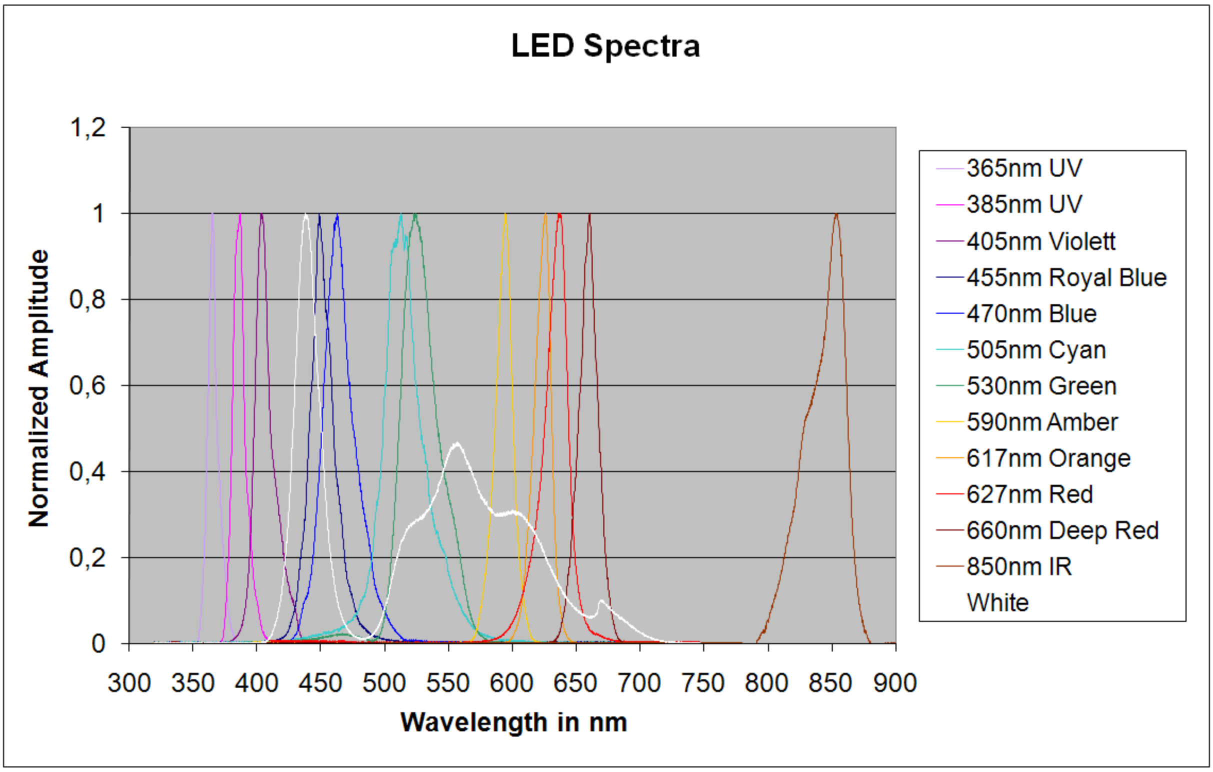
4.11 M850L1

Wavelength (nominal)	850nm
Optical Beam Power (LED Output)	400mW
Optical beam power (Collimated Beam)	120mW
Maximum LED Current	1000mA
LED Forward Voltage (nominal)	2.4V
Lifetime	100,000 hours

4.12 MCWHL1

Wavelength (color temperature)	5500K
Optical Beam Power (LED Output)	500mW
Optical beam power (Collimated Beam)	171mW
Maximum LED Current	700mA
LED Forward Voltage (nominal)	6.8V
Lifetime	500 hours

4.13 LED Spectra



5 Addresses

For technical support or sales inquiries, please visit us at www.thorlabs.com/contact for our most up-to-date contact information.

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